

CYTOLOGY

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Cytology: is a specialized branch of biology, which deals with the study of cells.

Cell

It is the smallest structural and functional unit in a multicellular organism.

1- Prokaryotic cells (pro = before, karyon = nucleus); are small, single - celled organisms (e.g. bacteria) that lack a nuclear envelope, histones, and membranous organelles.

2- Eukaryotic cells (eu = good) are cells with a true nucleus and exist as components of multicellular organisms.

Cellular components

1- Cytoplasm

2- Nucleus

1- Cytoplasm

The cytoplasm is composed of a matrix in which several structures are embedded. These structures are classified into:-

- Organelles**
- Cytoskeleton**
- Inclusions**

Cytoplasmic organelles

- They are organized structures located in the cytoplasm having a metabolic functions and are essential to the life of the cell.
- They are divided into:

A. Membranous organelles

- Cell membrane
- Mitochondria
- Rough endoplasmic reticulum
- Golgi apparatus
- Lysosomes
- Peroxisomes
- Endosomes
- Smooth endoplasmic reticulum

Cytoplasmic organelles

B. Non-membranous organelles

- Ribosomes
- Centrosome

Cell membrane (Plasmalemma)

- very thin semi-permeable membrane limiting the cell from its external environment.
- composed of **lipids** (phospholipids and cholesterol) and **proteins** with some **oligosaccharides** linked to some of the lipids and proteins.
- not visible with the light microscope because its thickness is about **7.5 nm**.
- The amount of lipids and proteins are approximately **equal**.
- By electron microscope (EM), the cell membrane appears as trilaminar (three - layered) structure. 2 dark lines separated by light one. Each layer is about 2.5 nm in width.

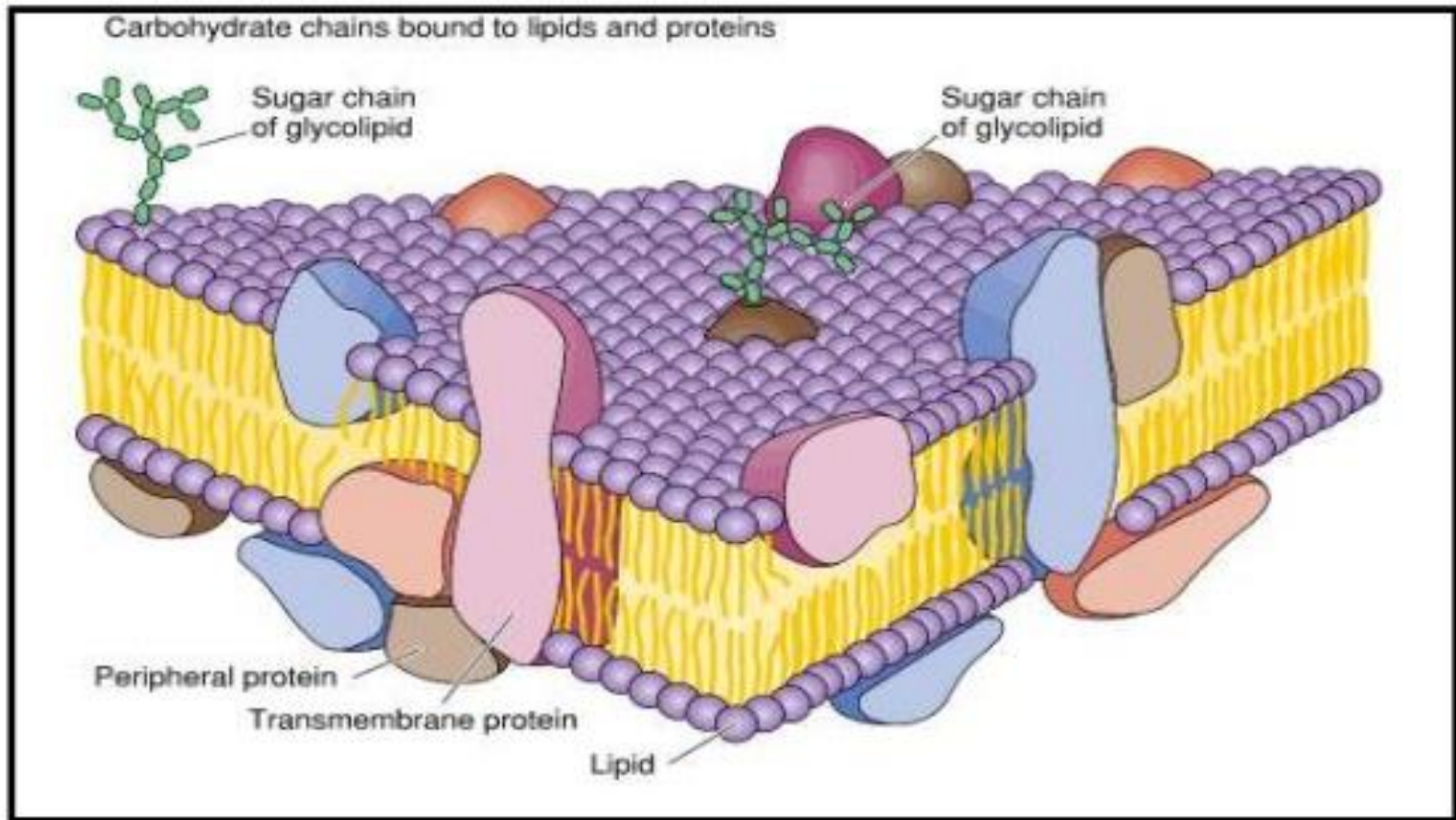
Cell membrane (Plasmalemma)

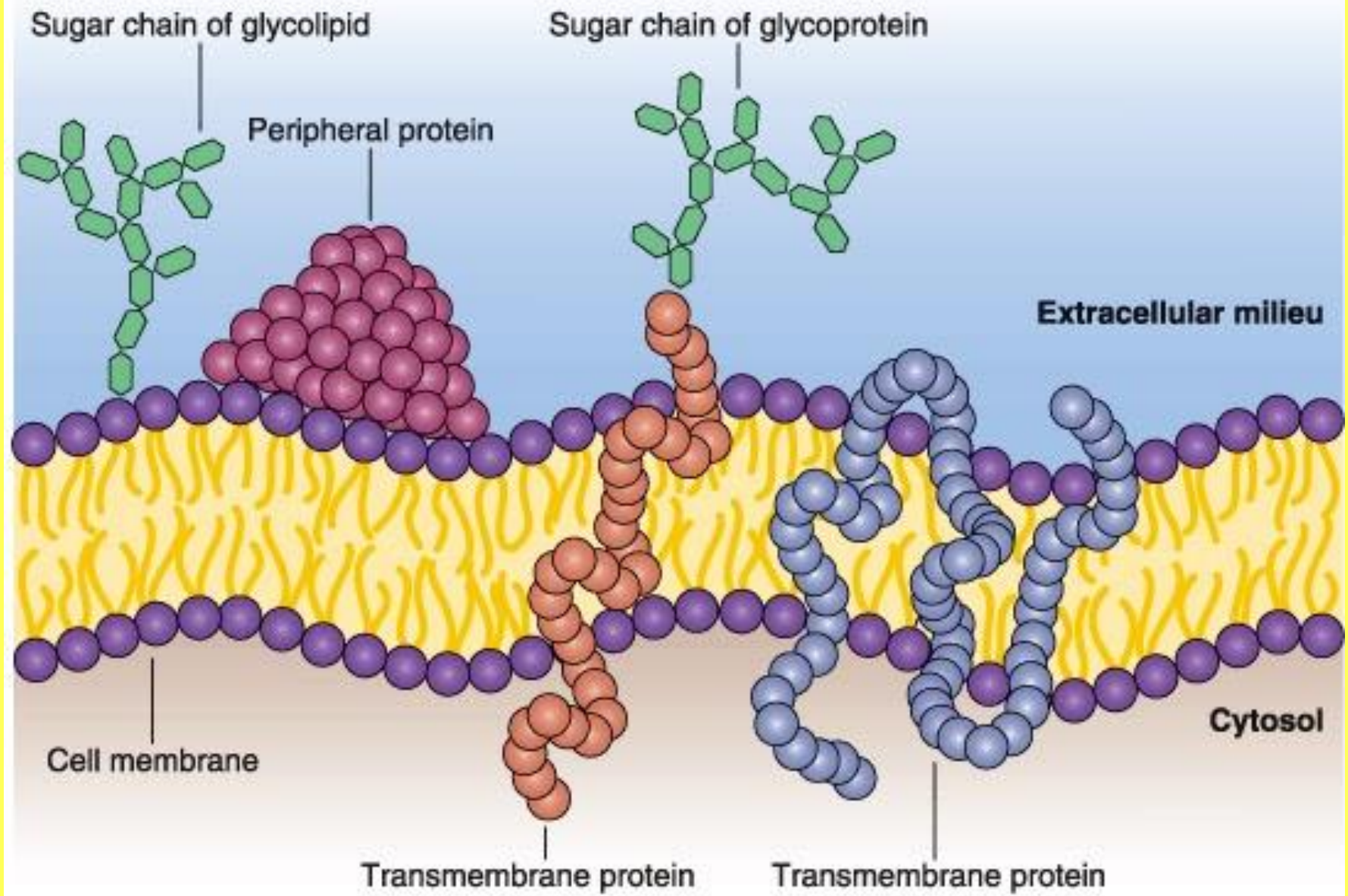
- ❑ The molecular arrangement of the cell membrane called fluid-mosaic-model.
- ❑ composed of bimolecular layer of phospholipid in which protein molecules are embedded
- ❑ Each phospholipids molecule is composed of a **polar head** (hydrophilic portion) located at the surface of the membrane (directed outward) and two long **non-polar tails** (hydrophobic portion) projecting into the center of the membrane (directed inward).

Cell membrane (Plasmalemma)

- ❑ The **protein molecules** are arranged in two groups
- ❑ **integral proteins** or **transmembrane proteins** which extend all the way from one side of the membrane to the other side.
- ❑ **peripheral proteins** which are attached to either aspect of the lipid bilayer (mainly to the cytoplasmic aspect) .
- ❑ This molecular arrangement of the cell membrane called **fluid-mosaic-model**

Diagram of Molecular Structure of Cell Membrane





Plasma Membrane Structure

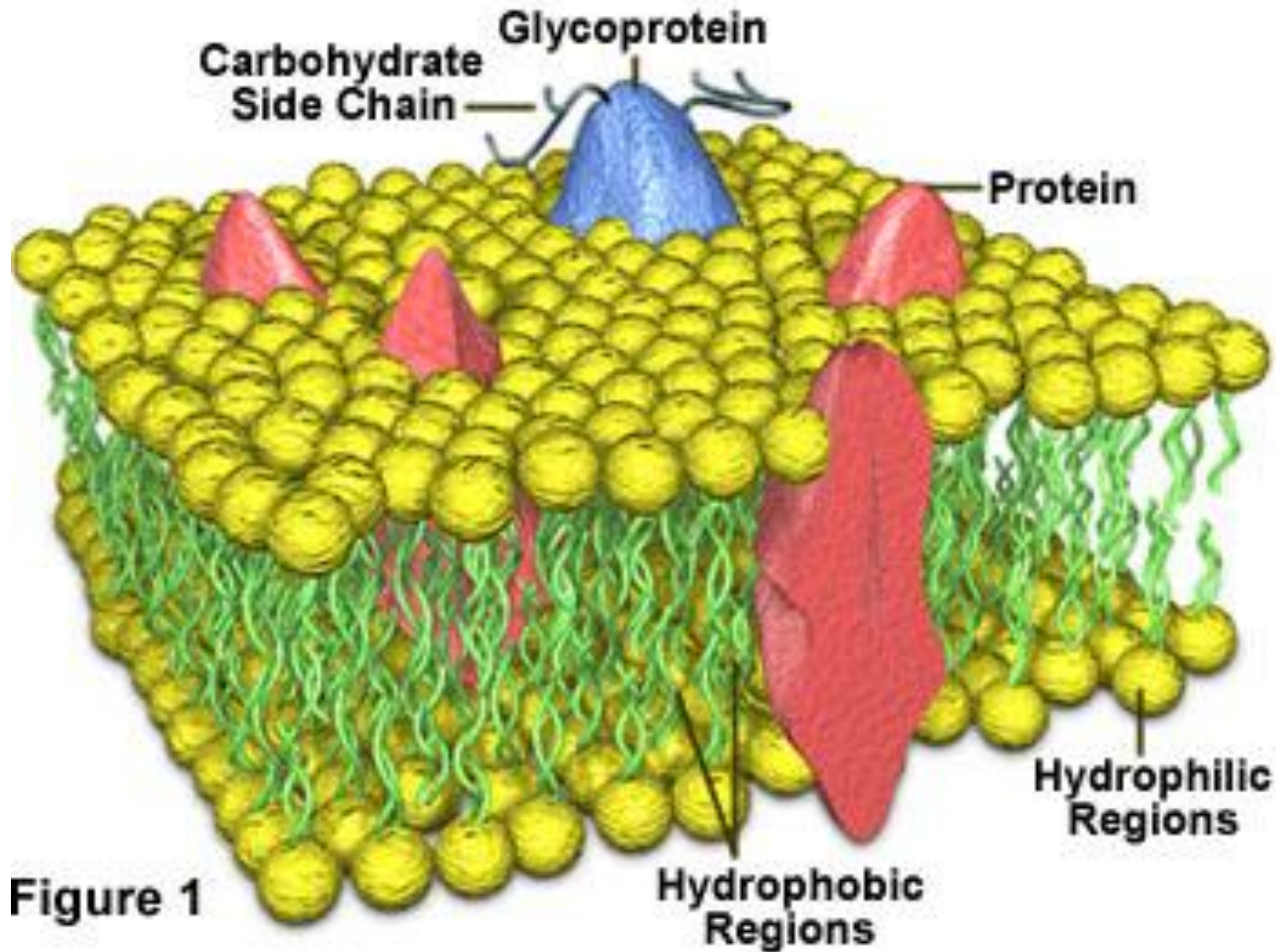
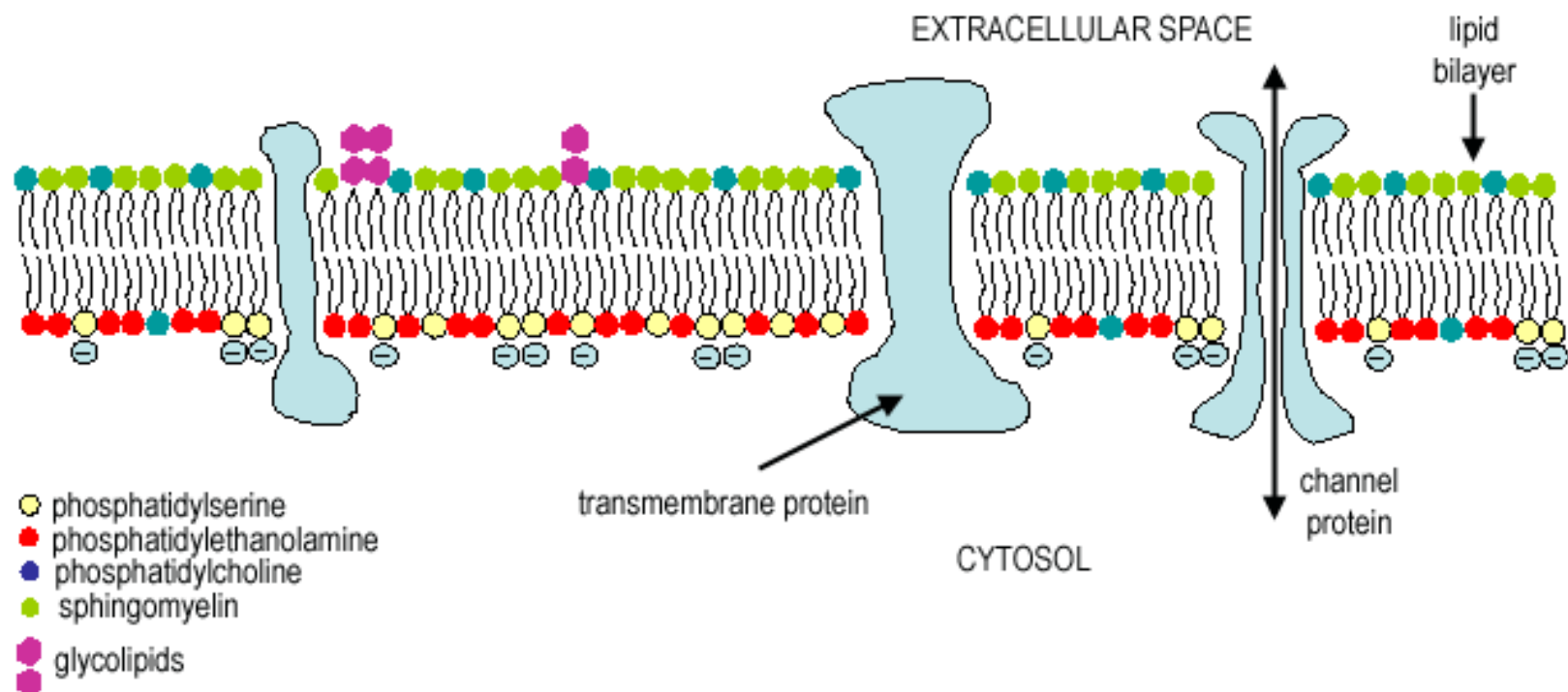


Figure 1



Cell membrane (Plasmalemma)

Glycocalyx

Many of the membrane proteins and some of the membrane lipids are linked with short chain of polysaccharides (**glycoproteins and glycolipids**) forming outer coat.

Note: - The **transmembrane proteins** make **channels** for the passage of certain ions and molecules in aqueous solution.

(the lipid molecules pass easily through the lipid-bilayer).

Cell membrane (Plasmalemma)

Functions:

1- Selective permeability

2- Some of the integral proteins on the surface are specific recognition molecules called cell-surface receptors.

3- The glycocalyx is involved in, cell recognition phenomena, intercellular adhesion, absorption and protection.

4- Endocytosis

a- Phagocytosis

b- Pinocytosis

c- Receptor-mediated endocytosis

5- Exocytosis

Cell membrane (Plasmalemma)

Membrane trafficking (flow):

- It is the movement of membrane from one organelle to another.
- Membranes bud as vesicles from an organelle and fuse with another membrane allowing the change of this membrane without membrane synthesis or breakdown.

Mitochondria

(mitos=thread, chondros=granule)

- It is a large organelle about 0.5 - 1 μm in diameter and 5-10 μm in length having variable **shapes**: ovoid, spherical or thread-like.
- **By EM** it appears bounded by **2 unit-membranes** each one is about **7 nm** thick separated by narrow **intermembranous space** (10-20 nm).
- The **outer** membrane has **smooth** contour, highly permeable that possesses a large number of **porins, multipass transmembrane proteins** each of which form a large aqueous channel.

Mitochondria

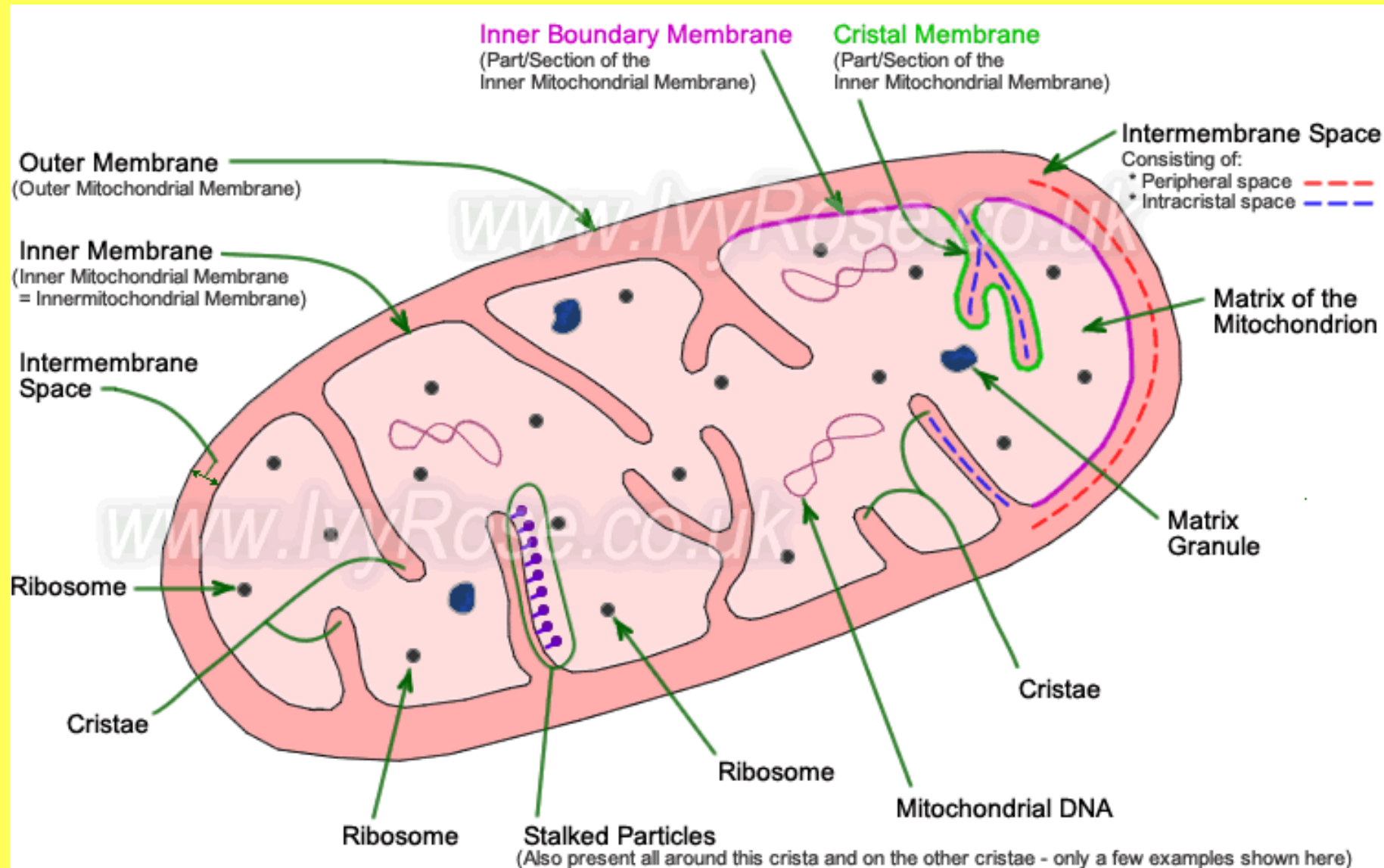
- The inner membrane is folded into **cristae** to provide a large surface area for **ATP synthase** and the **respiration chain**.
- The inner membrane is rich of **cardiolipin** (phospholipids) which makes it having high selective permeability.
- The cristae differ in number, size and shape from cell to another. They **increased** in cells with **high energy requirements**.

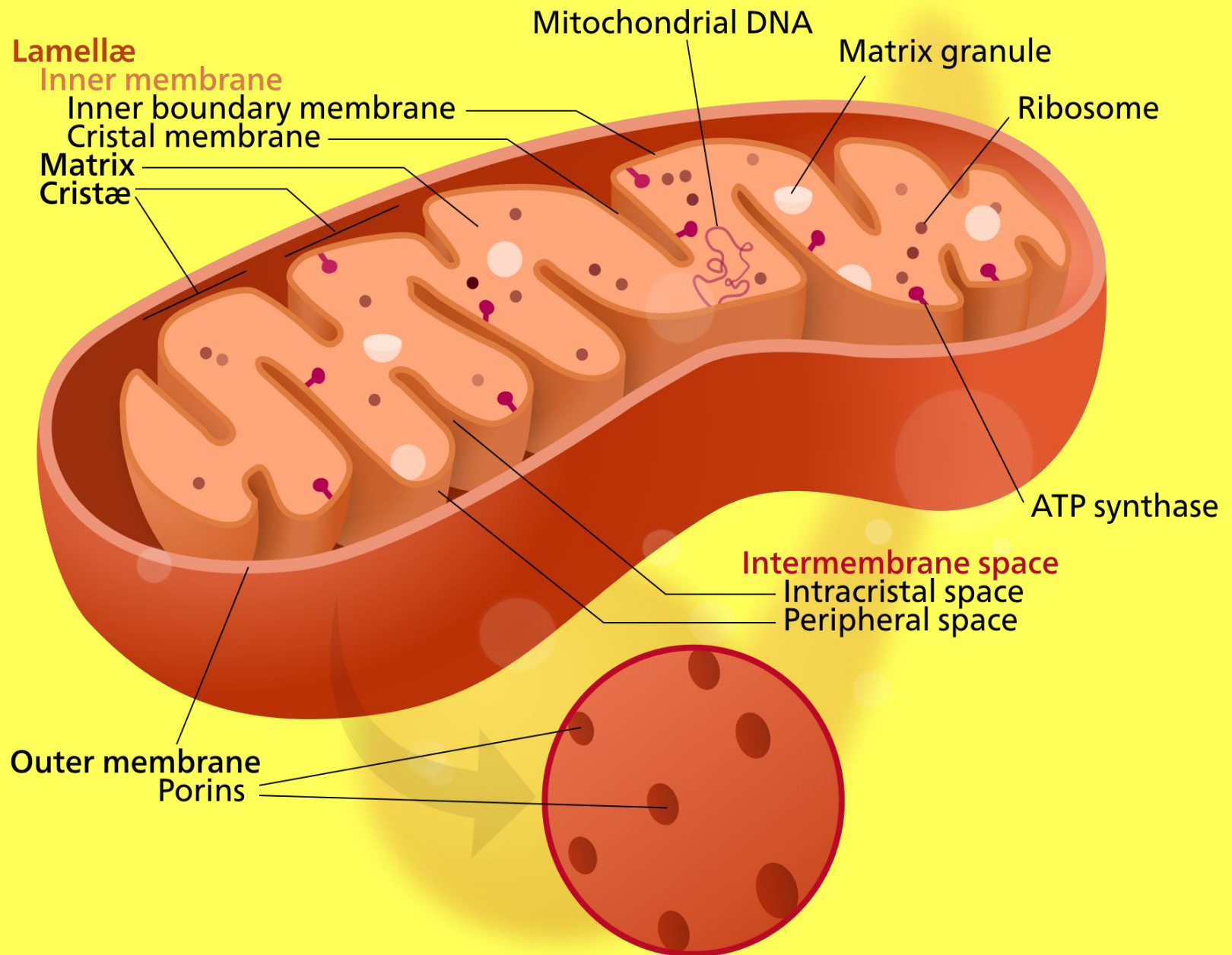
Mitochondria

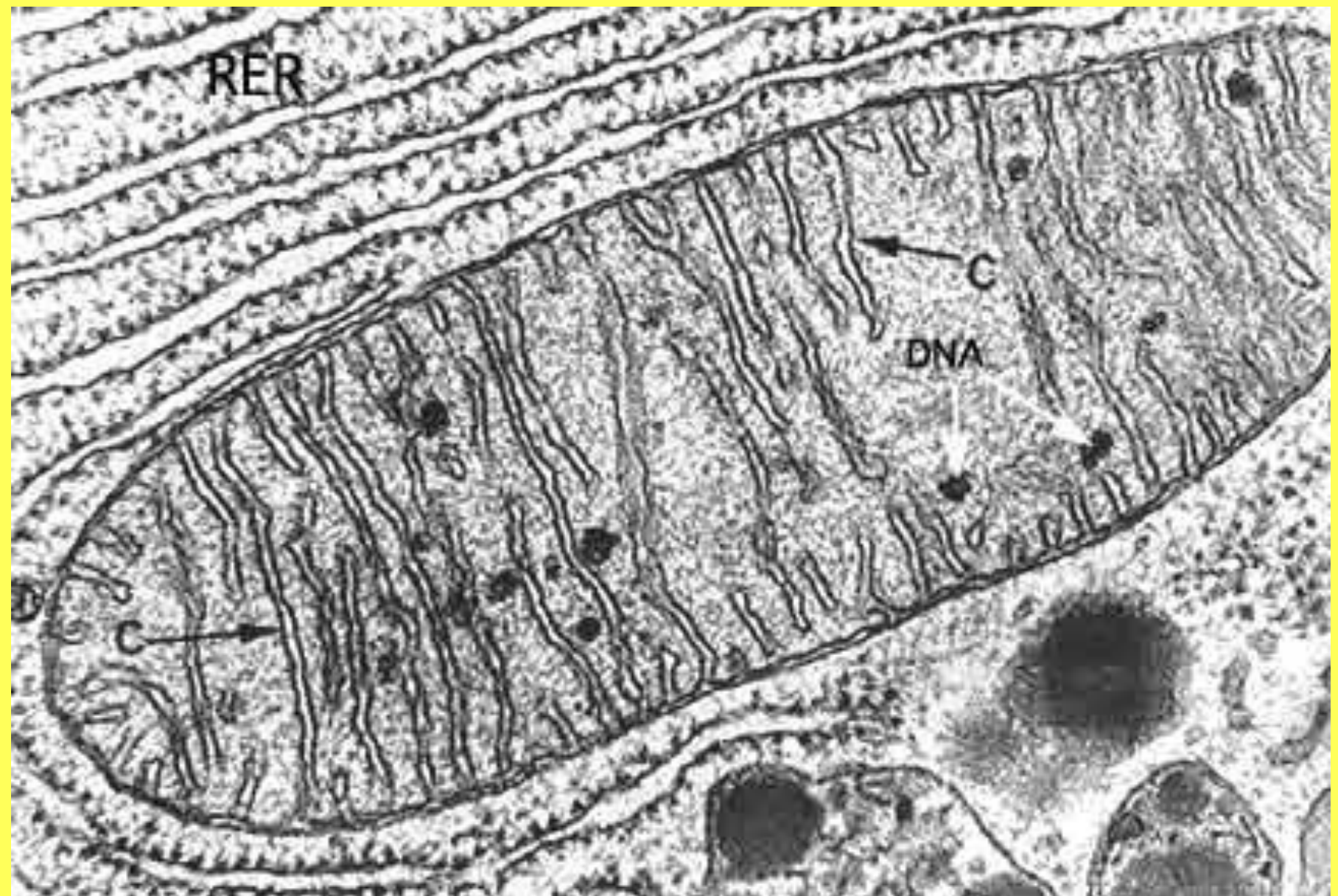
- The inner surface is covered by inner membrane **subunits**.
- Each subunit has a **globular head** 10nm in diameter attach to a narrow stalk projecting from the inner membrane into the matrix space.
- These subunits are **protein complexes** known as ATP synthase which generate ATP from ADP and inorganic phosphate.
- A large number of protein complexes (**respiratory chains**) are present in the inner membrane.

Mitochondria

- The mitochondrial matrix contains **water**, **solutes** and large dense granules (**cations** of some metals as calcium).
- It also contains strands of **DNA** and the three types of **RNA**.
- Many soluble **enzymes** are found having roles in the mitochondrial functions as the **Krebs cycle**, **B-oxidation of lipids** and **DNA synthesis**.
- Mitochondria can divide by **simple binary fission**.







Mitochondria

Functions:

- 1- It provides most of the **energy** needed for the cell.
- 2- It house many **enzymes** as those needed for the citric acid (Krebs) cycle and the fatty acid β -oxidation.
- 3- It concentrates cations of some metals e.g. **calcium**.

Rough endoplasmic reticulum (rER)

- It is anastomosing network of flattened sacs (**cisternae**) and tubules, their limiting membranes are studded by ribosomes.
- It is sometimes continuous with the outer membrane of the **nuclear envelope**.
- **Transfer vesicles**, are small membrane-limited vesicles that bud from the rER and cross the cytoplasm to reach the Golgi complex for further processing or packing.

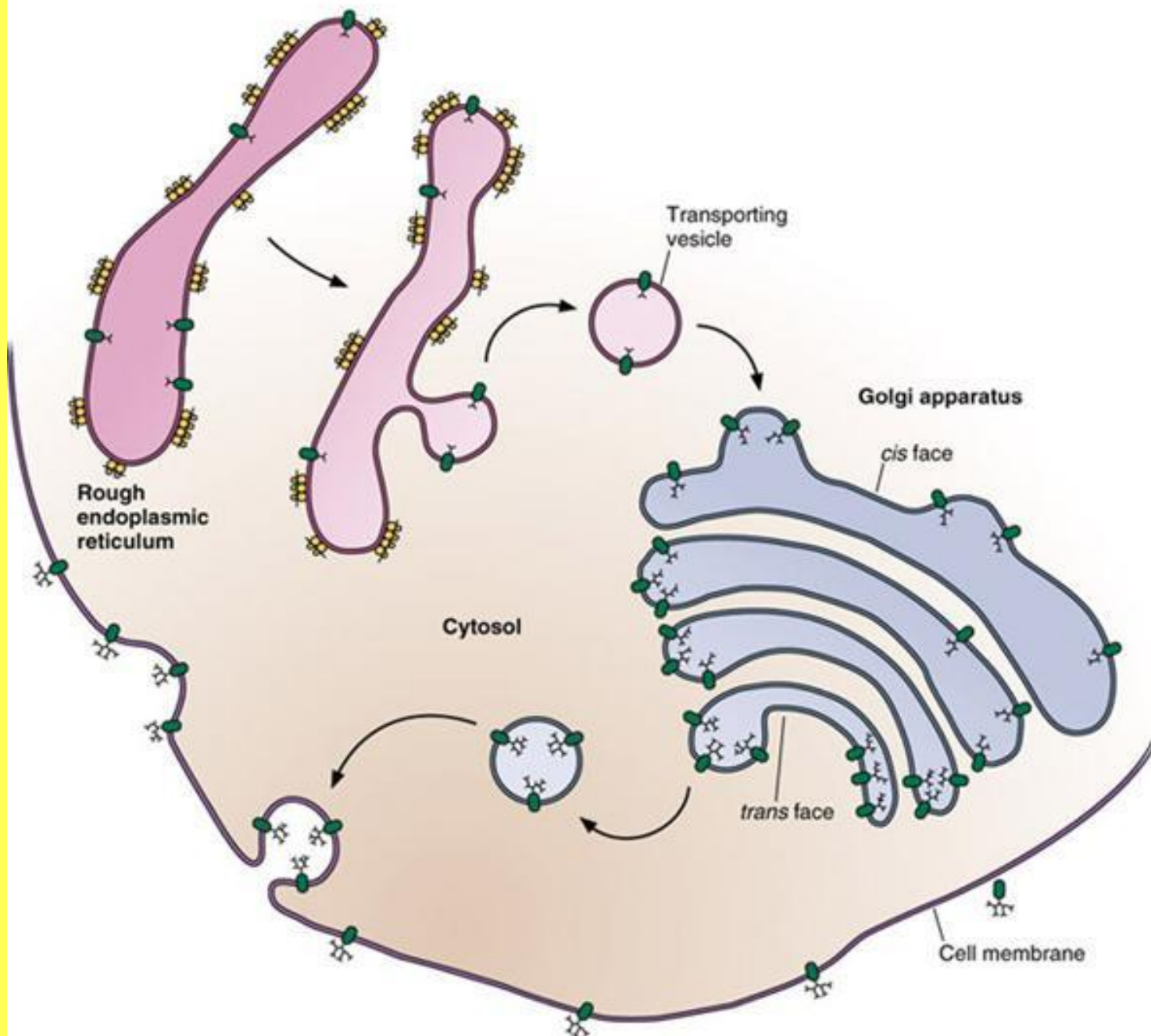
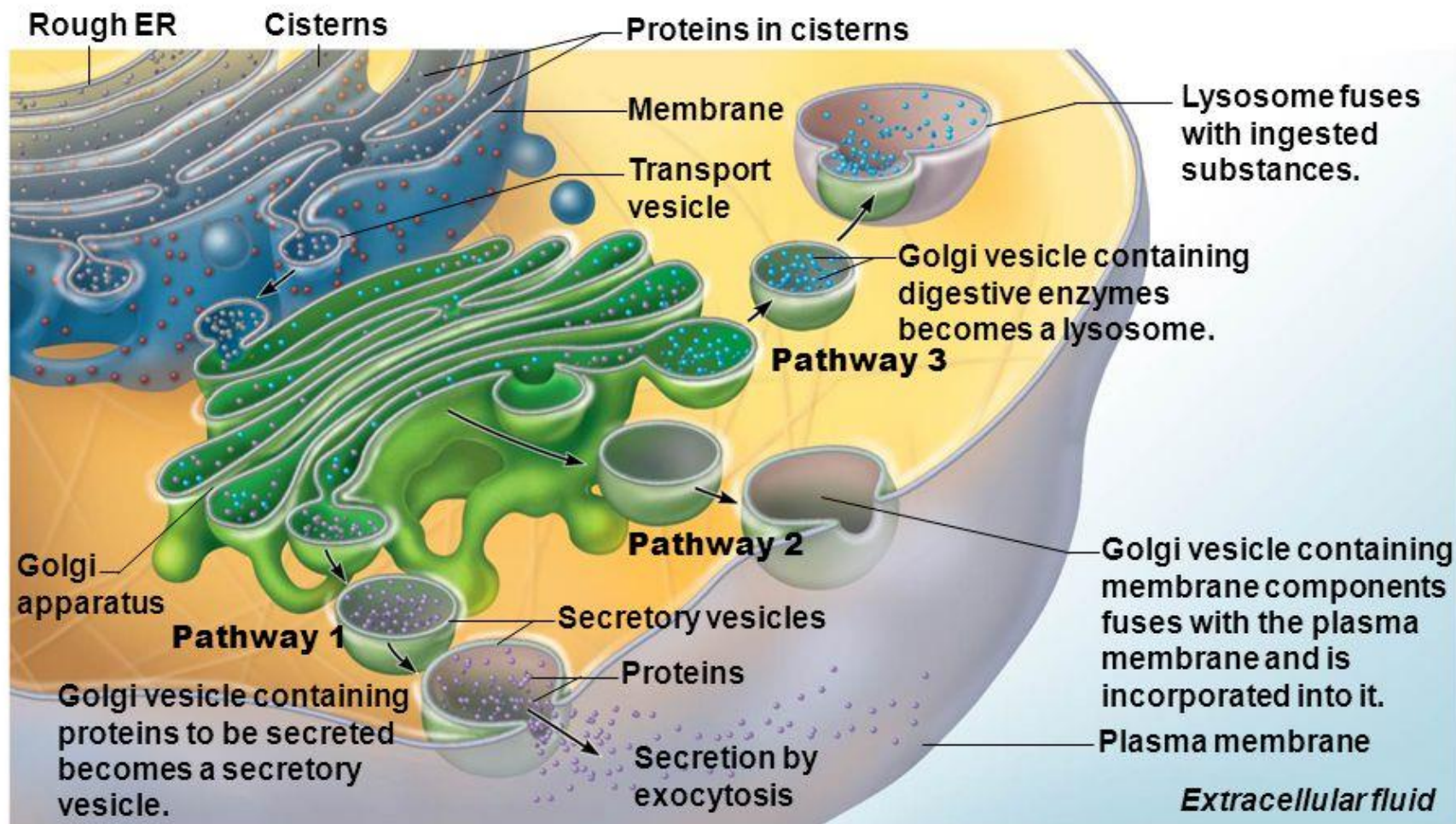


Figure 3.6 Role of the Golgi apparatus in packaging the products of the rough ER.



Rough endoplasmic reticulum (rER)

Functions:

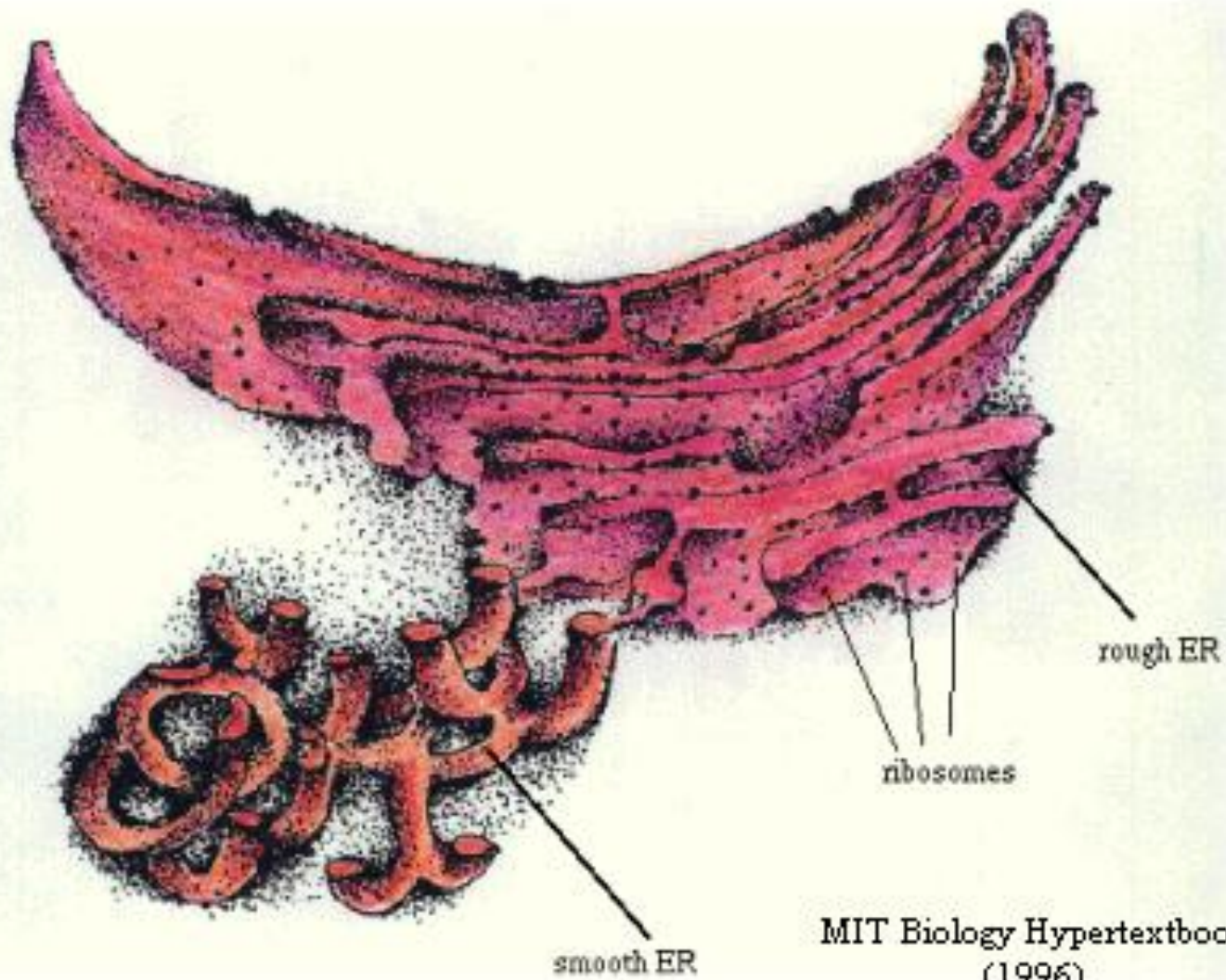
- 1- **Synthesis** and segregation of **proteins** that are eventually secreted from the cell.
- 2- Synthesis and segregation of **lysosomal enzymes**.
- 3- Formation of **integral membrane proteins**.
- 4- Modification of the synthesized proteins as sulfation, folding and glycosylation.

Smooth-endoplasmic reticulum:

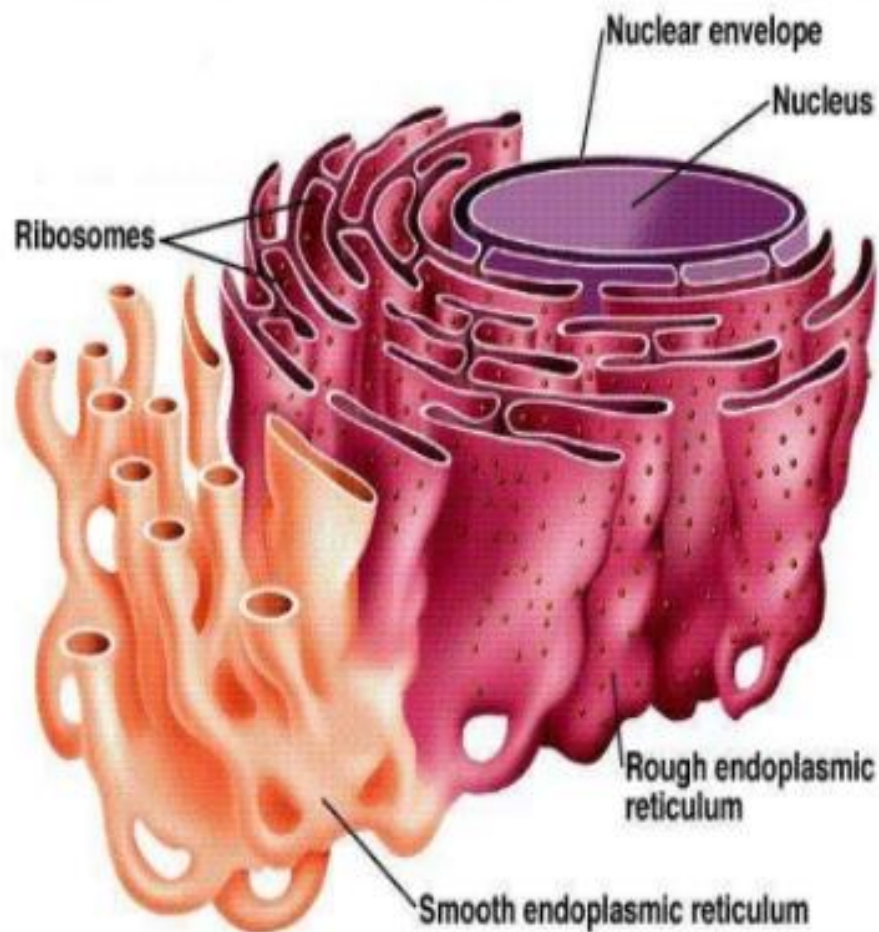
- It is a network of tubules that branch and anastomose in an irregular manner.

- It differs from the rER in that:
 - 1- Its limiting membrane is smooth (has no ribosomes).
 - 2- Its cisternae are more tubular than those of the rER.

- It may continue with rER or Golgi app.
- and its membrane **lacks** the integral membrane proteins.



MIT Biology Hypertextbook
(1996)

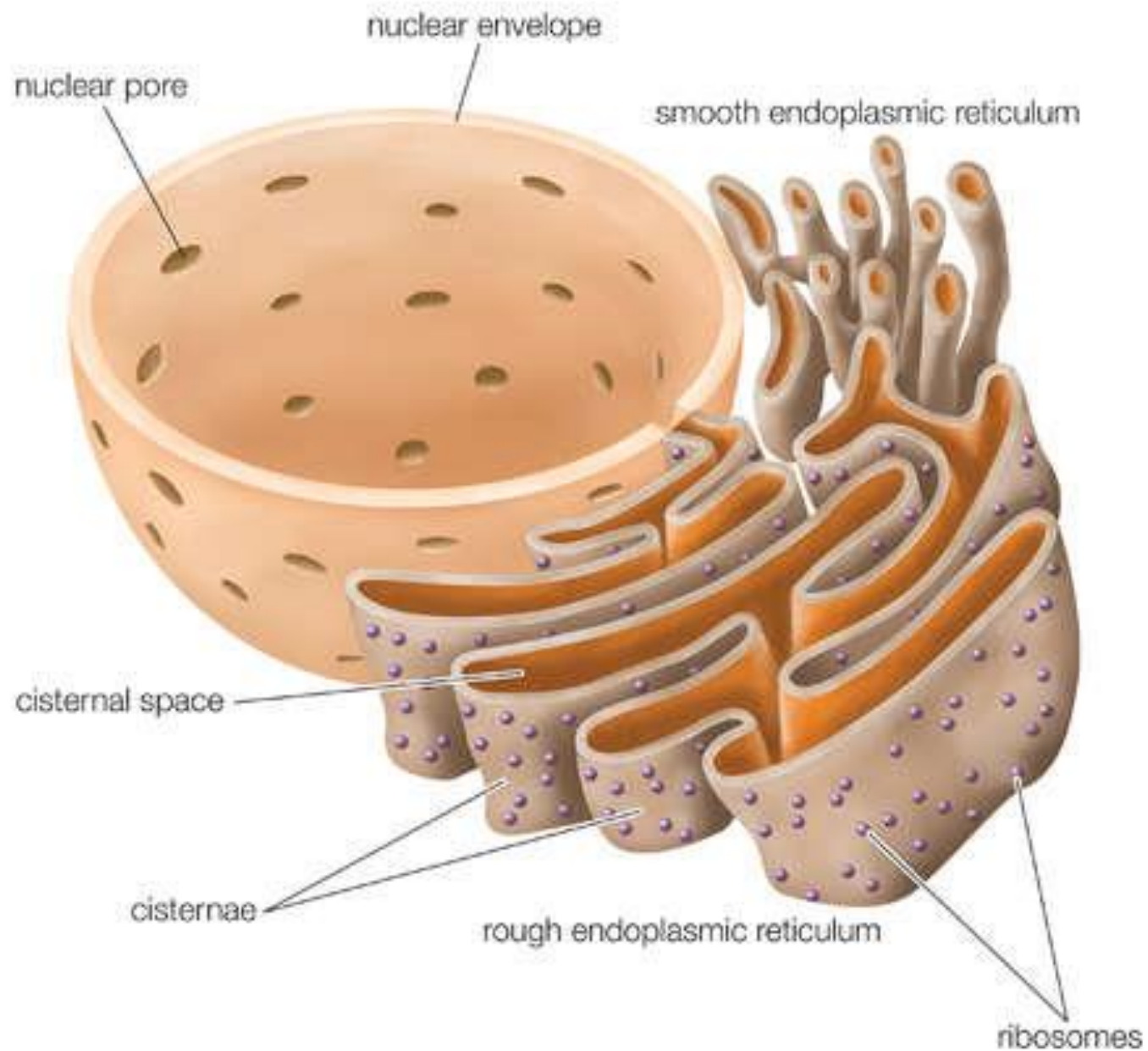


Рибосом сууж байрласан
эндоплазмын торыг *мөхлөгт
эндоплазмын тор*
(reticulum endoplasmicum
granurlosum) гэнэ.
Энд эсийн дотоод хэрэгцээний
болон эсээс гадагшлах
уураг, фермент нийлэгжиж
байдаг.

Харин рибосомтой
холбогдоогүй эндоплазмын
торыг *гөлгөр эндоплазмын тор*
(reticulum endoplasmicum
nongranurlosum) гэнэ.
Тэрээр өөх тос, нийлмэл нүүрс
усны солилцоог явуулах гол
хэсэг болдог.

Элэгнойр булчирхайн эсэд
мөхлөгт эндоплазмын тор их
байхад булчингийн эсэд гөлгөр
эндоплазмын тор нь илүү
давамгайл байдаг.

Endoplasmic reticulum



Smooth-endoplasmic reticulum:

Functions:

- 1- It is the site of intracellular **synthesis** of **lipids** and **cholesterol** derivatives (lipid, lipoprotein and steroid hormones).
- 2- It detoxify drugs.
- 3- It is involved in hepatic **glycogen metabolism**.
- 4- It regulate the intracellular distribution of **calcium** ions as in muscle cells.

Golgi apparatus (Gologi complex)

- It is a membranous organelle formed of 3 compartments:

- 1- A stack of 3-10 flattened sacs (cisternae) stacked together like a saucer.

- The stock of saccules has:

- Cis-face.

- Intermediate face (medial)

- Trans – face.

Golgi apparatus (Gologi complex)

- The cis-face is convex in shape, the newly formed proteins of the **transfer vesicles** which come from the rER enter it.
- The trans-face is concave in shape and is considered to be the exit face for the modified package proteins.

Golgi apparatus (Golgi complex)

2- Endoplasmic reticulum/Golgi intermediate compartment (ER/GIC) or Cis-Golgi network.

- It is associated with the **cis-face** and consists of vesicles and tubules formed from the fusion of the **transfer vesicles**.

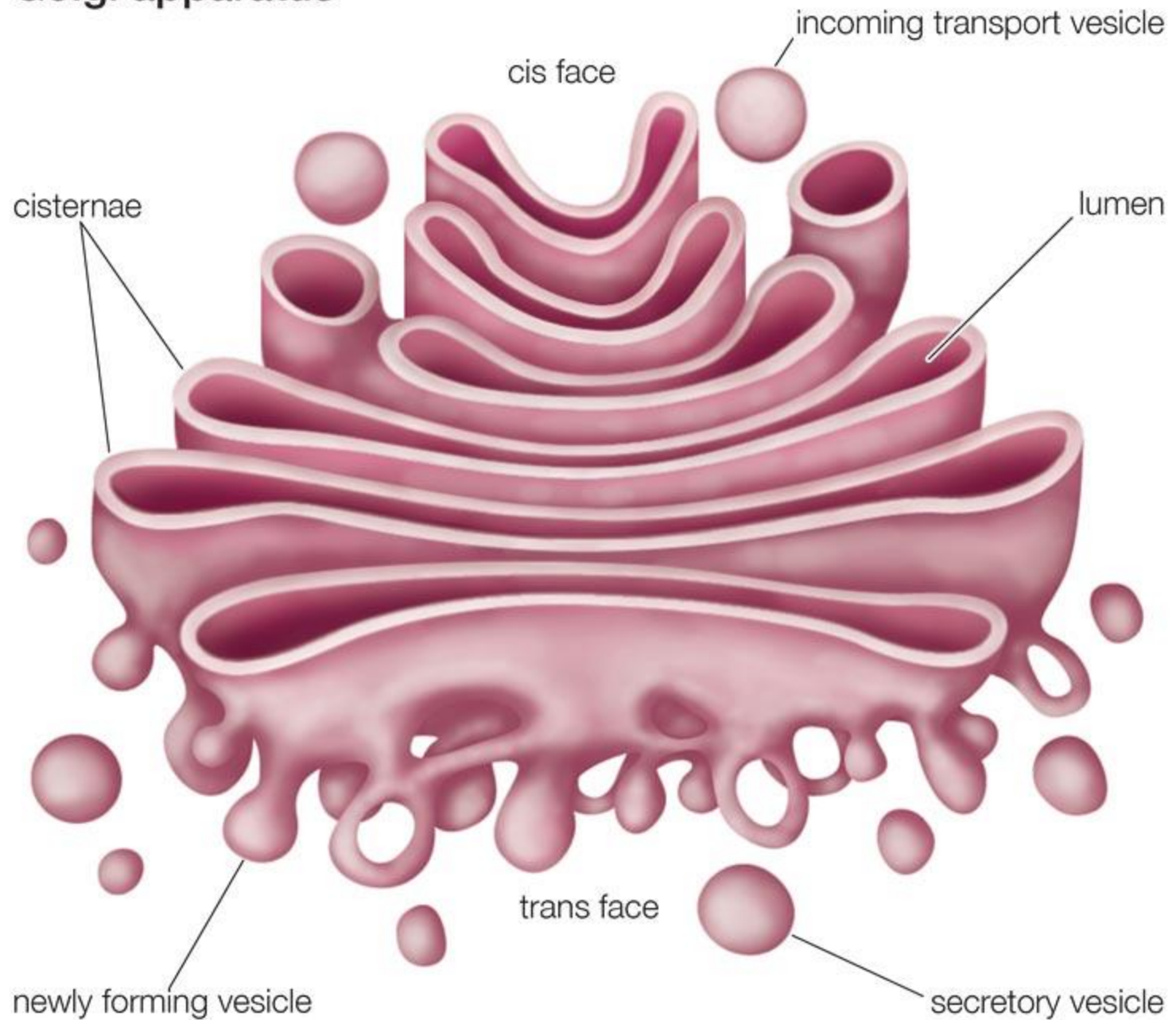
3- Trans-Golgi network

- Located at the distal side of the Golgi apparatus and is responsible of sorting of proteins to their pathways, so that they reach the cell membrane, **secretory granules** or **lysosomes**.

Golgi apparatus (Gologi complex)

- ❑ The proteins are transferred from the cis-face to the trans-face via **vesicles** that bud off and fuse with the rim of the compartments.
- ❑ Within the different compartments, the proteins modify.

Golgi apparatus



Golgi Apparatus

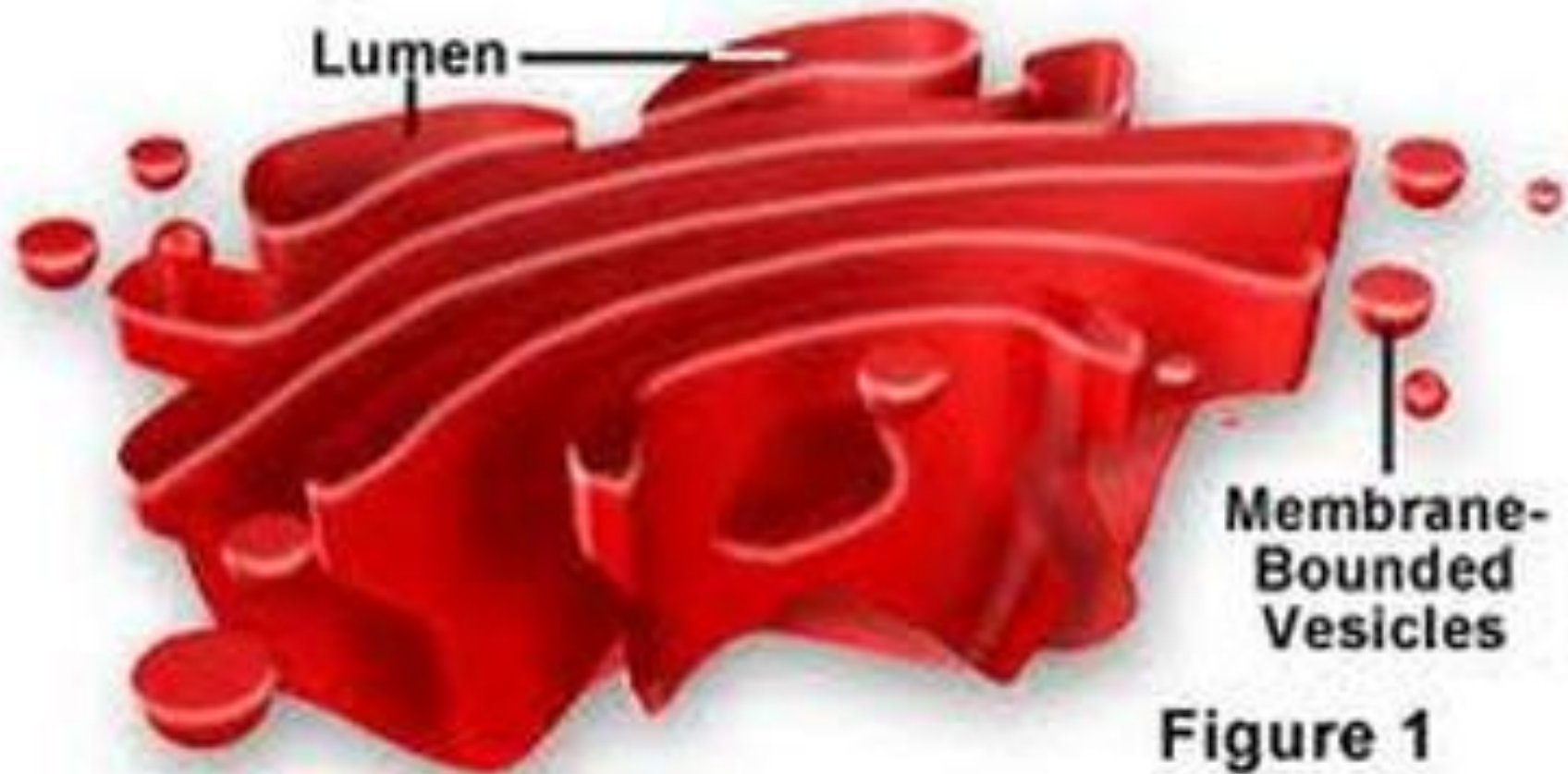


Figure 1

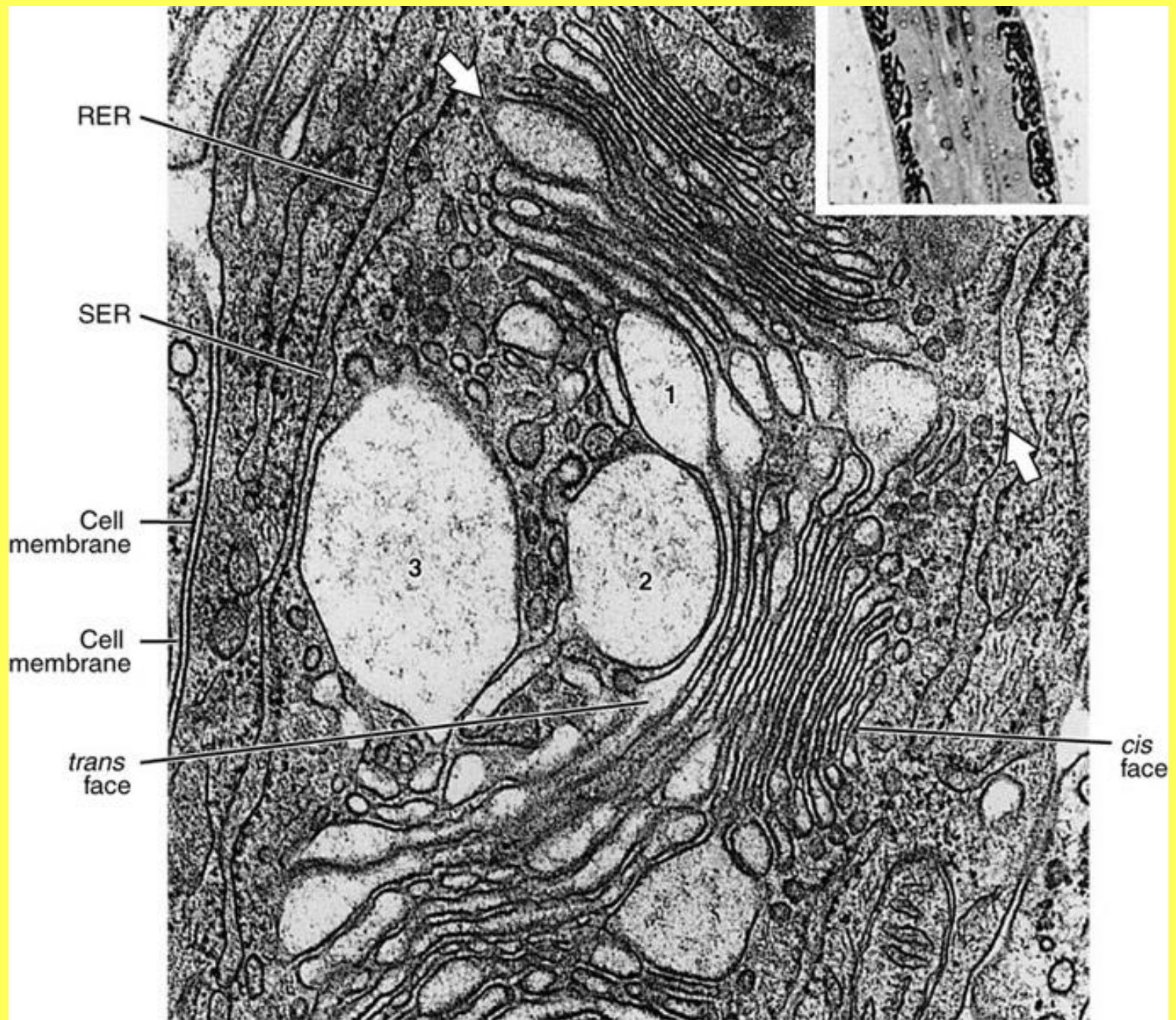


Figure 3.6 Role of the Golgi apparatus in packaging the products of the rough ER.

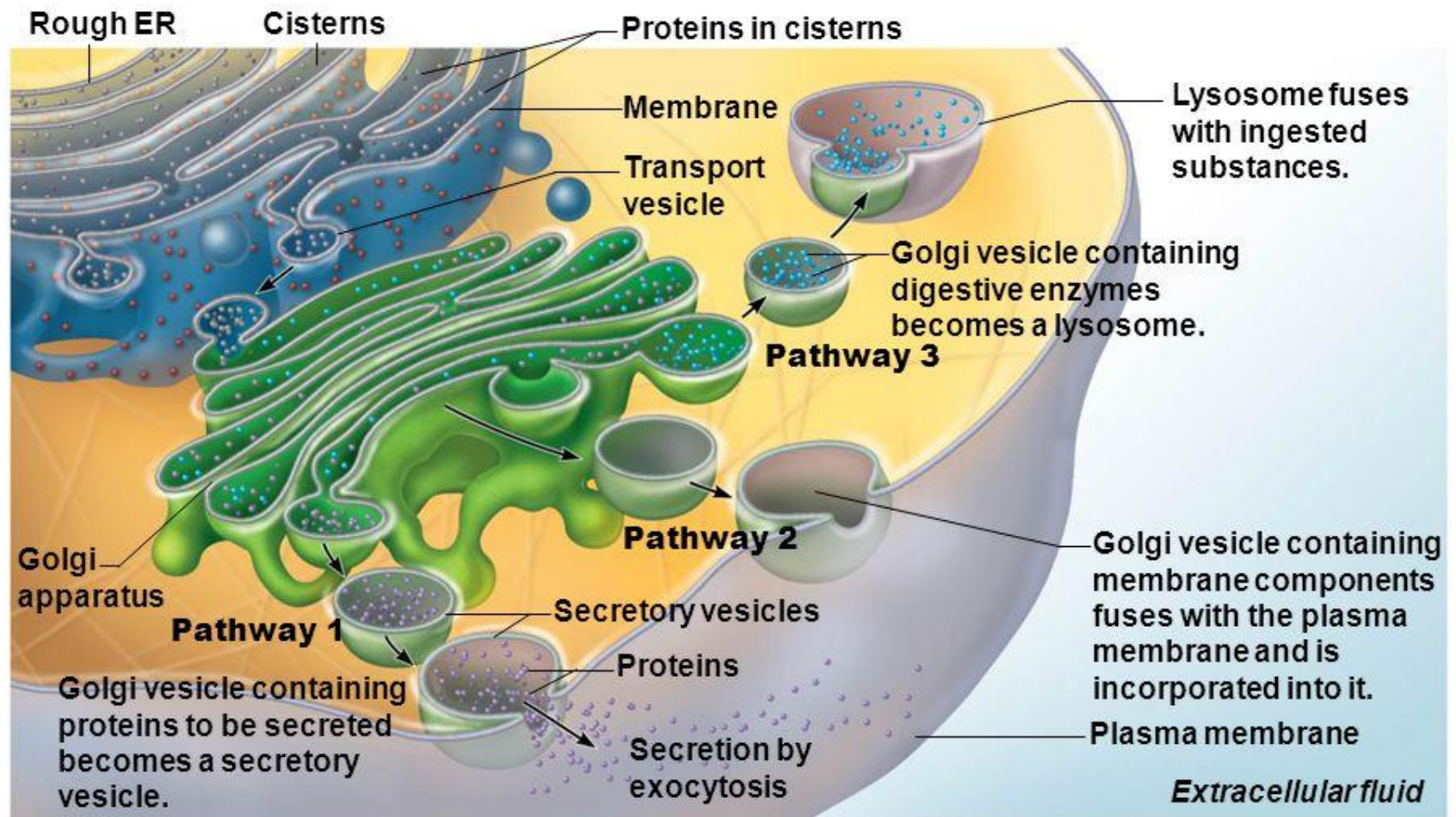
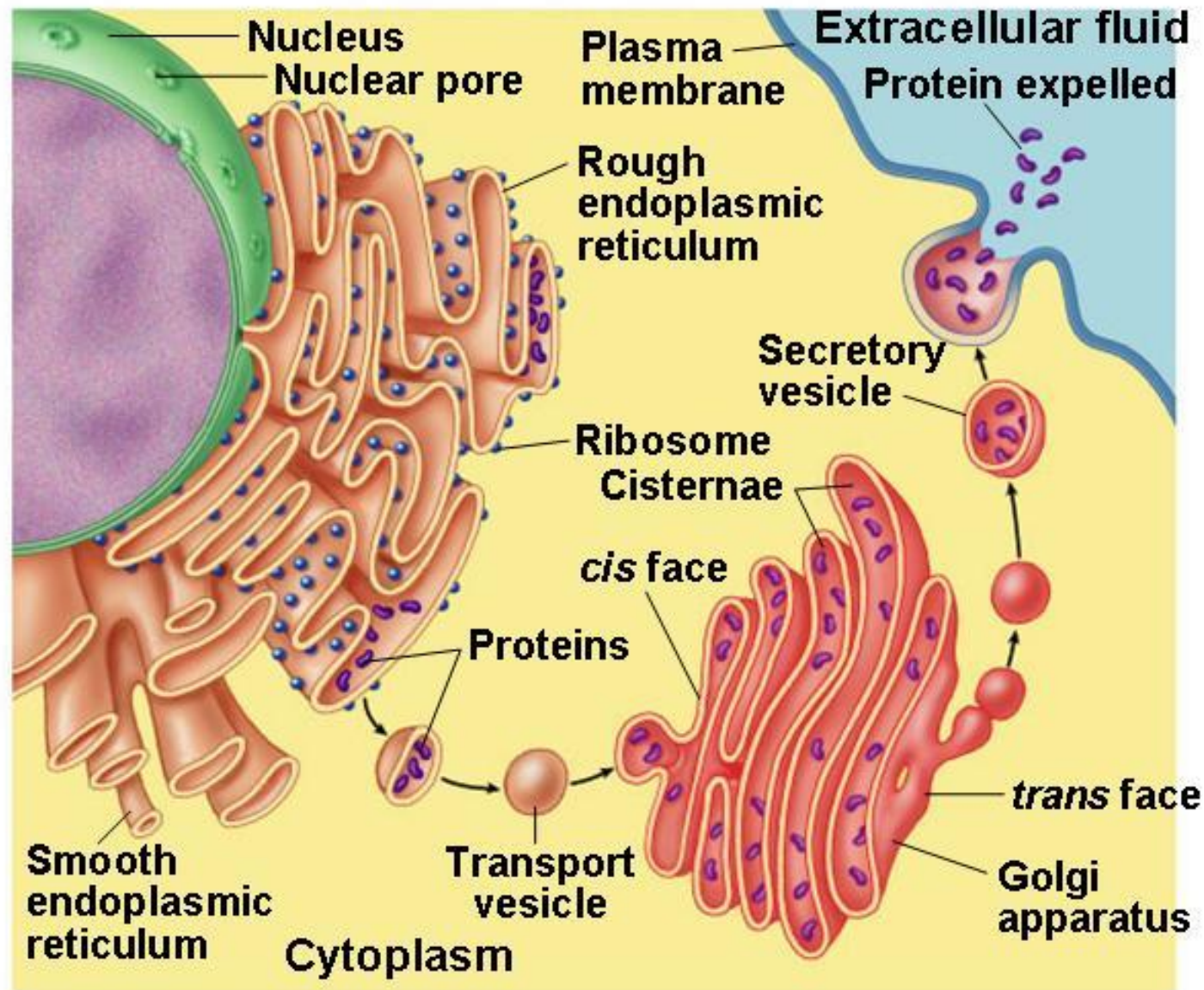


Fig. 5.16 (TEArt)

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Golgi apparatus (Gologi complex)

Functions:

- 1- **Ripening** and **modification** of the secretory products .
- 2- It **segregates** the acid hydrolyses in a membranous organelle called lysosome.
- 3- It is the main center of **membrane traffic** within the cell, directing and redirecting newly synthesized constituents as integral glycoprotein and recycling membranes.
- 4- It has a role in both **lipoprotein secretion** and **prohormone processing**.

Lysosomes

- They are spherical membrane-limited vesicles about 0.05-0.5 μm in diameter.
- contain large numbers of **hydrolytic enzymes** and function as the **cellular digestive system** (site of intracellular digestion and turnover of cellular components).
- These enzymes are more than 40 enzymes as acid phosphatase, lipases, glucosidases, proteases, nucleases, are formed in rER transferred to Golgi app. and packed as lysosomes.

Lysosomes

- Lysosomes arise from the **trans-Golgi network** covered by a **clathrin** coat which is lost then the uncoated vesicle fuse with late **endosomes**.
- The lysosomal enzymes are bounded to receptors which **dissociate** within the **endosome** and the receptors are recycled and returned to the trans-Golgi network.
- They vary in size and density, depending on their level of activity:

Lysosomes

1- Primary lysosomes:

- small (5 - 8 nm in diameter) having electron dense contents.
- They are the storage form of lysosomes and appear uniform granular in EM.

2- Secondary lysosomes:

- larger and less electron dense having **mottled** appearance in EM.
- They are formed by fusion of one or more primary lysosomes with a phagosome.
- Following digestion of the contents, nutrients diffuse through the lysosomal membrane and enter the cytoplasm.

Lysosomes

- The undigestible substances are retained within the vacuoles (**residual bodies**).

3- Residual bodies:

- membrane-limited inclusions of various sizes and densities associated with the terminal phases of lysosome function.
- They contain **indigestible** materials as pigments, crystals.
- Some cells as **macrophages** expel the residual as waste, but long-lived cells as **nerve cells** tend to accumulate them (**lipofuscin pigment**).

Lysosomes

Lysosome Structure

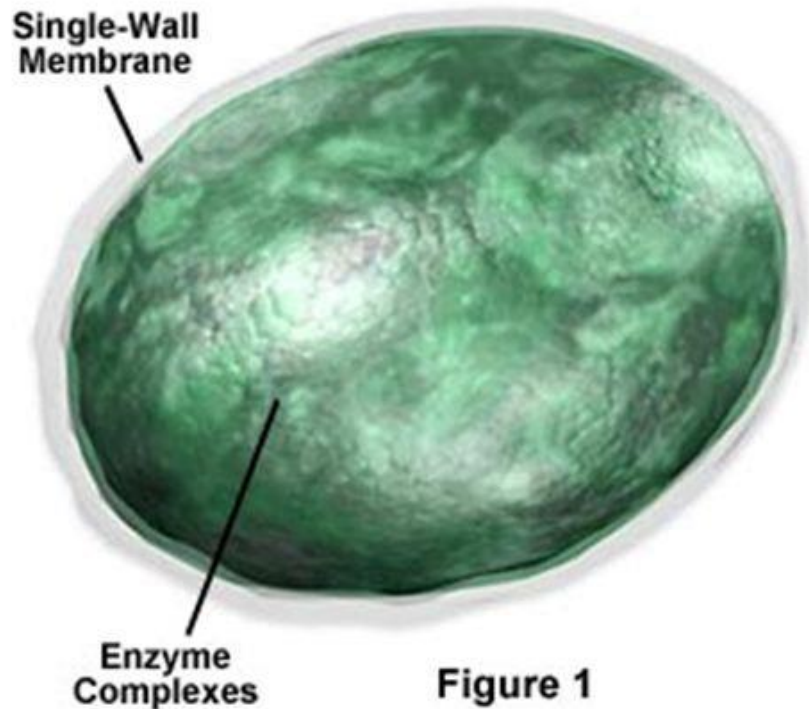
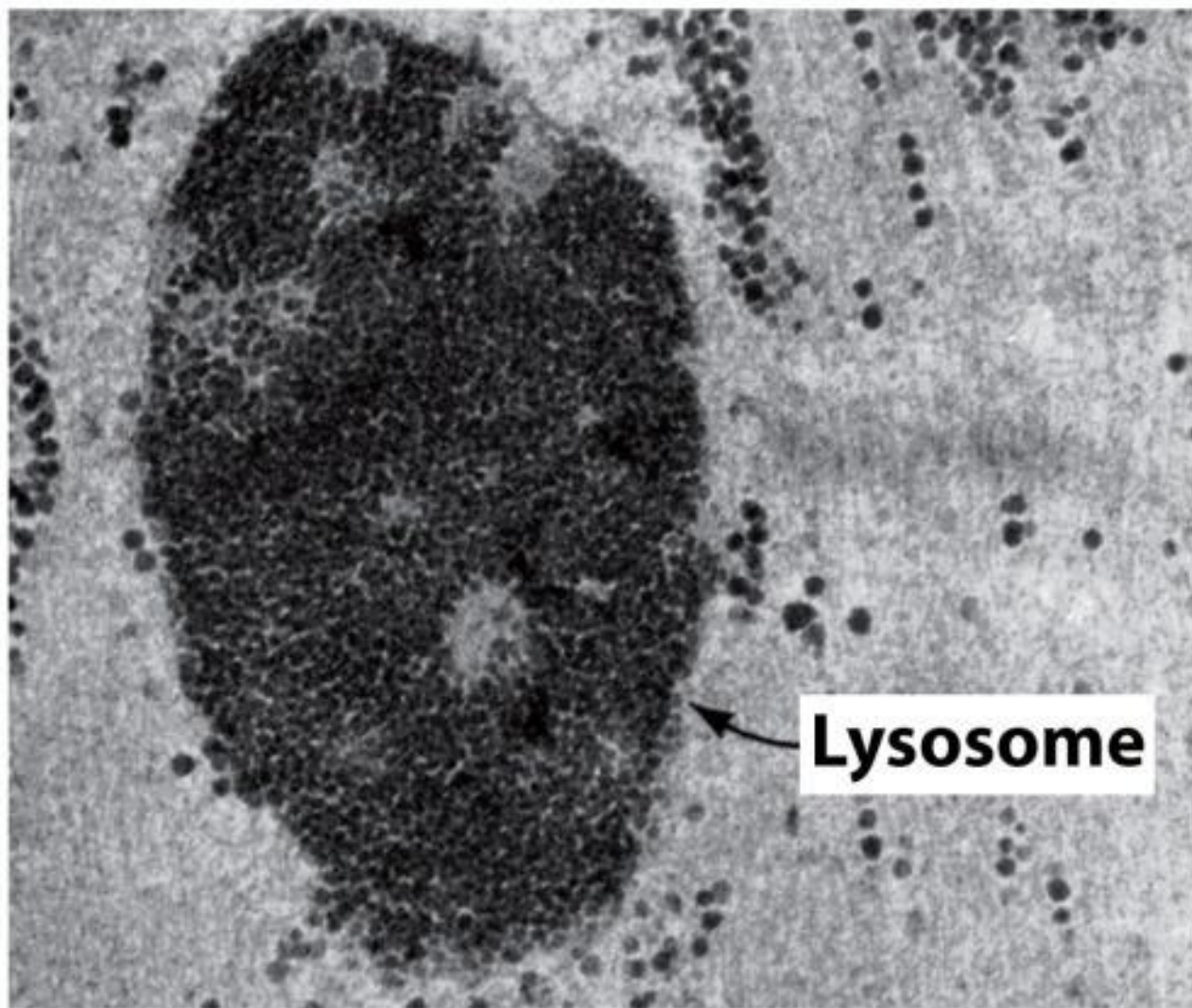


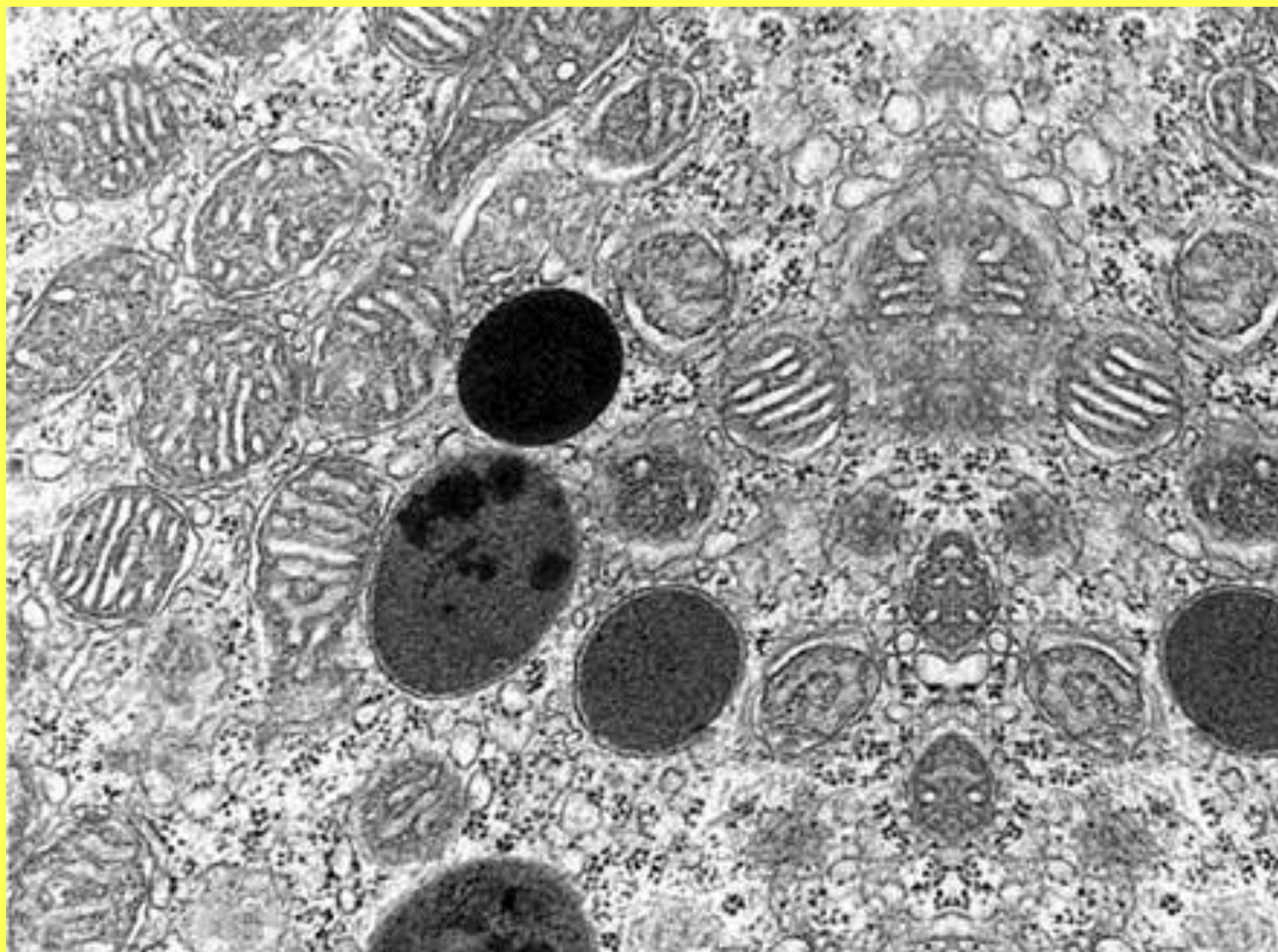
Figure 1

- Recycling Center
 - Recycle cellular debris
- Membrane bound organelle containing a variety of enzymes.
- Internal pH is 5.
- Help digest food particles inside or outside the cell.

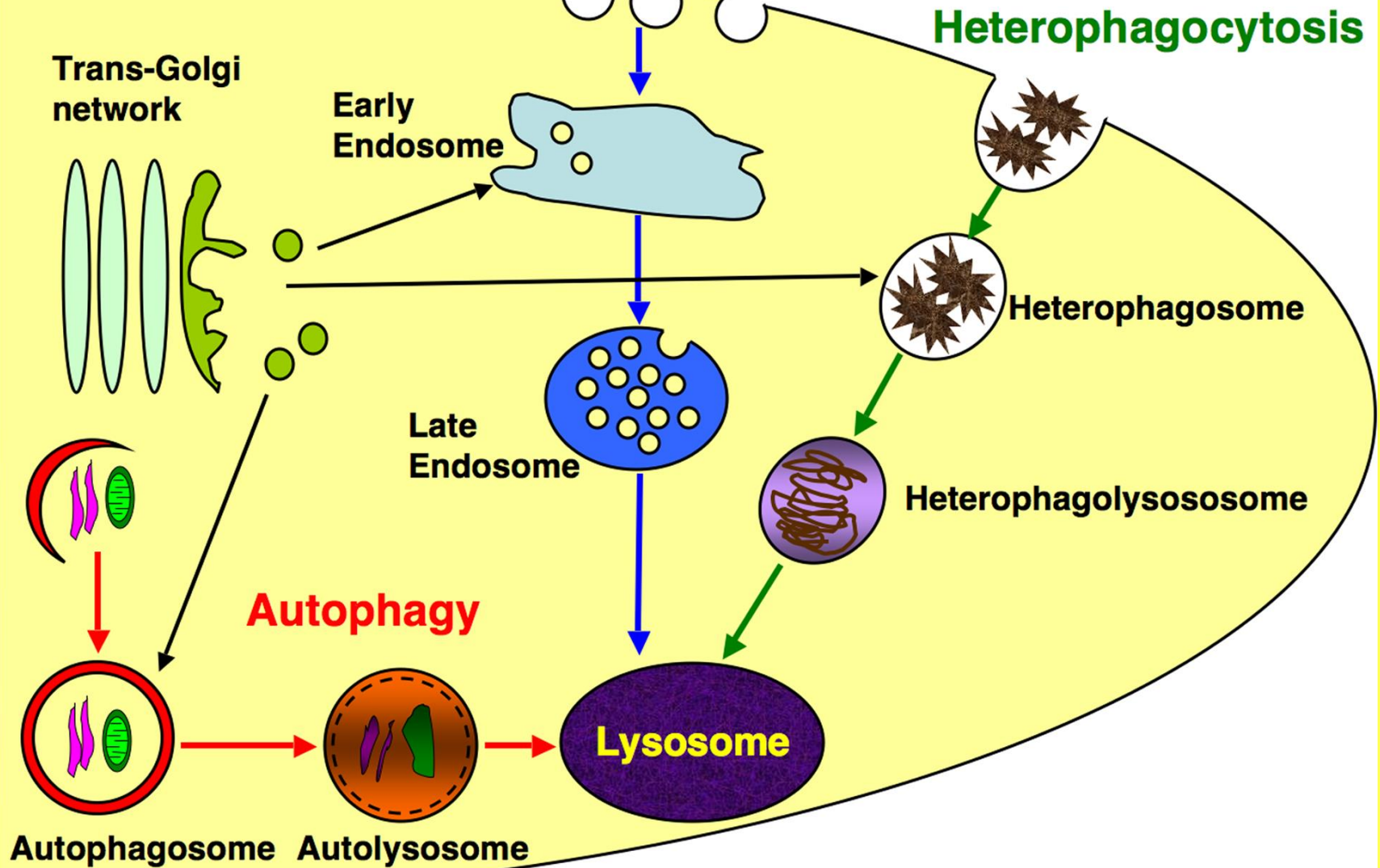


1 μm

Figure 21.25
Biochemistry, Seventh Edition
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Endocytosis



Lysosomes

Functions:

- 1- It degrades any exogenous particles that enter the cell (**heterophagy**).
- 2- It dispose any cellular constituents or organelles that are no longer useful to the cell (**autophagy**).
- 3- Hypoxia (drop in oxygen tension) causes escape of the lytic enzymes to the cytoplasm (**autolysis**).
- 4- It play an important role in the **metabolism** of several substances and the deficiencies of its functions cause diseases as the storage disease of **glycogen** in the liver cells.

Peroxisomes (microbodies)

- They are membrane-limited enzymes-containing vesicles **larger than the primary lysosomes** and varies in shape,size and enzyme content which form a population of **tiny vesicles**.
- In many species it has central crystalloid structure called a **nucleoid** (uricase enzyme that degrades urates).
- Not found in **man** and **birds**.

Peroxisomes (microbodies)

- They called peroxisomes because they contain **peroxide forming enzymes** and **catalase** and are involved in the formation of **hydrogen peroxide** which is used by phagocytic cells to kill ingested microorganisms and in certain metabolic reactions as also detoxifies various noxious agents e.g. ethanol .
- It has a role in **lipid metabolism** and the **production of glucose from lipid**.

Peroxisomes (microbodies)

- The **oxidase** enzymes produce **hydrogen peroxide** and the **catalase** converts it into **oxygen** and **water**.
- They grow in size and **divide** by fission into two smaller ones whoever they possess no genetic material for its own.

Endosome:

- A system of vesicles and tubules located in the cytoplasm **near** the cell surface (**early endosome**) or **deeper** in the cytoplasm (**late endosome**).
- It's membrane contains ATP-driven H^+ pump that acidify its interior.
- It is about 200-400 nm in diameter and has **vesicular to tubular shape**.
- It **receives coated vesicles** (ligands + receptors) after **shedding** their coats (**Clathrin**), then **dissociation** occurs, thus the **receptors return** to the **cell surface** to be used again and the **ligands** transferred to **late endosome**.

Endosome:

- The content of the late endosome enter lysosome for digestion.

N.B.

Coated vesicles need surface receptors to pick up certain substances (ligands) needed by the cell .
(about 20 different macromolecules are know to enter the cell by this process).

Proteasomes:

- They are small organelles composed of **protein complexes** that are responsible for **proteolysis of malformed proteins**.
- Proteins as those of the metabolic regulation have to be degraded to ensure that the metabolic response to a signal stimulus is not prolonged.
- Also **denaturated, damaged or malformed** proteins have to be eliminated.
- the process of proteolysis is carefully controlled by the cell.